Survival Models April 2023 Solutions

1. (a) Survival function calculation:

Event Time	n_j	d_j	$\frac{n_j - d_j}{d_j}$	$\hat{S}(t)$
0^-	15	0	1	1
1^-	15	1	14/15	0.933
6^-	12	1	11/12	0.856
11^-	10	1	9/10	0.770
13^-	8	1	7/8	0.674
17^-	2	1	1/2	0.337

[4]

(b) We assume that the right-censoring is non-informative. [1]

This may not hold in this example if a patient leaves the study due to being too ill to attend a doctor's appointment. [1]

(c) $\hat{S}(t)$ is only defined for $t \leq 19$ since all we know about the observation censored at time $t = 19$ is that they were still alive at the end of the study.

[1]

[Total: 7 marks]

2. (a) A cross-sectional study involves following a number of mini-cohorts (e.g. ages $x \to x+1$) over a relatively short period of time (e.g. 3 years). [2]

Each age range can be used to provide an estimate of $\hat{\mu}_x$ or $\hat{q}_x$ using a binomial model, Poisson model or Markov model for example. [1]

Using these estimates the data can then be graduated, for example using a mathematical formula or with reference to a standard table, to provide a smooth function of age. [1]

(b) We are mixing mortality since, for example, the mortality of a 60 year old now is not the same as that of a 57 year old in three years time. [1]

Cross-sectional studies introduce Type I censoring since most lives will not die during the observation period. [1]

(or any other sensible points)

(c) A model can built whereby μ or q is a function of both age and time, for example a Lee-Carter or Cairns-Blake-Dowd model.

This can then be projected into the future using a simle time series approach or similar.

[2 marks for any sensible description]

[Total: 8 marks]

3. (a)
- **Markov assumption:** The probability that a life now aged x will be in a particular state at any future time $x+t$ depends only on the age x and the state currently occupied. [1]
- For any two distinct states g and h the transition probability ${}_{dt}p_{x+t}^{gh}$ is given by
$$ {}_{dt}p_{x+t}^{gh} = \mu_{x+t}^{gh}\, dt + o(dt) $$
as $dt \to 0$. [1]
- The probability that a life makes two or more transitions in time dt is $o(dt)$ as $dt \to 0$. [1]
- The range of ages x is such that the transition rates can be assumed to be approximately constant, e.g. all values of x are similar. [1]

(b) The likelihood is given by
$$ L(\mu, \lambda) = e^{-\mu v}\mu^d \times e^{-\lambda v}\lambda^u $$
[1]

and so the log-likelihood is given by
$$ \ell(\mu, \lambda) = -\mu v + d\log\mu - \lambda v + u\log\lambda $$
[1]

where

v is the total observed waiting time in the working state,

d is the total number of deaths observed

and u is the total number of individuals leaving the company. [1]

(c) Taking the derivative of the log-likelihood with respect to λ gives

$$\frac{\partial \ell}{\partial \lambda} = -v + \frac{u}{\lambda}.$$

[1]

Setting this to 0,

$$0 = -v + \frac{u}{\hat{\lambda}},$$

gives

$$\hat{\lambda} = \frac{u}{v}.$$

[1]

(d) We approximate 6000 months as 500 years, noting that $\hat{\mu}$ is given as a rate per annum.

Since

$$0.01 = \hat{\mu} = \frac{d}{v} = \frac{d}{500}$$

we have $d = 5$. [1]

There are 40 employees so $40 - 5 = 35$ left the company. We therefore have

$$\hat{\lambda} = \frac{u}{v} = \frac{35}{500} = 0.07$$

per annum. [1]

[Total: 11 marks]

4. (a) Using the uniform distribution of deaths assumption, we have

$$ {}_{b_i}q_{70+a_i} = \frac{(b_i - a_i)q}{1 - a_i q}. $$

This gives

$$ \begin{aligned} L(q) &\propto \prod_{i=1}^{5} q_i^{d_i}(1-q_i)^{d_i} \\ &= q \times \left(1 - \frac{0.5q}{1-0.5q}\right) \times 0.5q \times \left(1 - \frac{0.5q}{1-0.25q}\right) \times \frac{0.75q}{1-0.25q} \\ &\propto \frac{q^3(1-q)(1-0.75q)}{(1-0.5q)(1-0.25q)^2}. \end{aligned} $$

[3]

Taking the logarithm gives

$$ \ell(q) = 3\log(q) + \log(1-q) + \log(1-0.75q) - \log(1-0.5q) - 2\log(1-0.25q). $$

[1]

(b) Taking the derivative with respect to q gives

$$ \frac{\partial \ell}{\partial q} = \frac{3}{q} - \frac{1}{(1-q)} - \frac{0.75}{(1-0.75q)} + \frac{0.5}{(1-0.5q)} + \frac{0.5}{(1-0.25q)}. $$

[1]

One can then check that

$$ U(0.7368) = 0.00045248 \approx 0 $$

implying that $\hat{q} = 0.7368$ is a maximum likelihood estimate. [1]

(c) We have

$$ I(q) = -\frac{\partial^2 \ell}{\partial q^2} = \frac{3}{q^2} + \frac{1}{(1-q)^2} + \frac{0.75^2}{(1-0.75q)^2} - \frac{0.25}{(1-0.5q)^2} - \frac{0.125}{(1-0.25q)^2}. $$

[1]

Using the estimate $\hat{q} = 0.7368$ above gives $I(0.7368) = 21.957$ [1]

and so the standard error is approximately $\sqrt{1/21.957} = 0.2134$. [1]

(d) We have a waiting time of

$$v = 0.5 + 0.5 + 0.25 + 0.5 + 0.375 = 2.125$$

[1]

and we have $d = 3$ deaths so

$$\hat{\mu} = \frac{d}{v} = \frac{3}{2.125} = 1.412.$$

[1]

(e) Assuming constant force of mortality, $\hat{\mu} = 1.412$ implies

$$\hat{q} = 1 - e^{-1.412} = 0.7563.$$

[1]

We see this is similar to the value from (b). In particular, an approximate 95% confidence interval for the estimate in (b) is given by

$$(0.7368 - 1.96 * 0.2134, 0.7368 + 1.96 * 0.2134) = (0.3185, 1.1551)$$

so we see the estimate $\hat{q} = 0.7563$ lies well within this confidence interval. [1]

[Total: 13 marks]

5. (a) (i) D could be a parameter which particularly affects high ages x while E could be a parameter which particularly affects low ages x. [2]

(ii) Relatively low amounts of data for low and high ages could lead to unreliable estimates of D and E. [1]

(b) Graduating against a standard table may be appropriate if:

- Information on a more extensive graduation of similar data is used.
- Little data is available, especially at extreme ages.
- You wish to incorporate prior expectations of the form of the curve.

[1 mark for every sensible answer]

(c) We may record the numbetr of claims made instead of the number of deaths, leading to incorrect estimates of death numbers. [1]

This can lead to increased variation in our data causing overdispersion in graduated mortality rates. [1]

[Total: 8 marks]

6. (a) Life year rate intervals are rate intervals where a change in age labels refers directly to a policyholder's birthday. [1]

A calendar year rate interval is where the age label is determined with respect to a fixed point in the calendar year. [1]

A key difference is that, unlike life year rate intervals generally, the number of lives exposed to risk in a calendar year interval is not a continuous function. [2]

(b) The Principle of Correspondence states that death data and exposure data must use the same age definition. [1]

That is, if a life is counted as being exposed to risk of death at age label x then if he or she were to die, he or she would be counted in the deaths at the same age label. [1]

(c) (i) Age next birthday [1]

since this definition is used in the death data [1]

[one can argue that it does not matter here]

(ii) $d_{45} = 32 + 33 + 34 = 99$.

Let $P^D_{x,t}$ be the number of lives aged x next birthday at time t.

Let $P^C_{x,t}$ be the number of lives aged x last birthday at time t.

Then $P^D_{x,t} \approx P^C_{x-1,t}$. [1]

This gives

$$\begin{aligned} E^c_{45} &= \int_{t=0}^{3} P^D_{45,t} dt \\ &\approx \int_{t=0}^{3} P^C_{44,t} dt \\ &\approx \frac{1}{2} \times 7645 + 8430 + 8224 + \frac{1}{2} \times 7331 \\ &= 24142. \end{aligned}$$

[1]

We then have

$$\hat{\mu}_{45} = \frac{99}{24142} = 0.0041$$

at exact age 44.5. [1]

(iii) We have

$$\hat{q}_{45} = 1 - e^{-\hat{\mu}_{45}} = 1 - e^{-0.0041} = 0.00409$$

[1]

applying at exact age 44 [1]

[The whole of Question 6.(c) can use the opposite conversion since there is an almost exact correspondence between age last birthday and age next birthday except at the ends of the interval]

[Total: 13 marks]